

Data In, or Privacy Out? Deploying OptBot: A Survey Tool to Assess Privacy Notice Design on Data Consent Willingness for AI Workflows

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Abstract

Privacy notices help people decide whether to share data, but long or dense notices can be hard to use at the moment of consent. This study compared two versions of the same simulated AI model-improvement notice: an icon-led visual layout and plain text. Each participant saw both notices in random order and rated willingness to share, trust, completeness, and ease of use. The final data set included 30 complete paired responses and no excluded records.

Willingness was slightly higher for the visual notice than for plain text (3.40 vs. 3.20 on a 1–5 scale). The average within-person difference was +0.20, with a 95% bootstrap confidence interval from −0.07 to 0.50. The median difference was 0, and an exact sign test was not statistically significant ($p = .424$). The extra outcome measures were mixed, and their confidence intervals also included zero. The study does not show that the visual design reliably increases willingness to share. It provides a careful survey tool and study plan that can be improved in a larger follow-up study.

1. Introduction

When people decide whether to share data for AI improvement, they may be deciding about prompts, files, assistant replies, feedback, and technical data. A notice needs to do more than list these facts. It needs to help people understand what they are agreeing to.

This project tested two ways of showing the same OptBot Assistant privacy notice. Notice A used an icon-led disclosure ledger. Notice B used the same information in plain text. The public OptBot repository documents the survey tool, its validation checks, randomized notice order, and the rule that participant data do not belong in the public repository (Candela, 2026). Appendices A–C reproduce the administered survey, consent screen, and both notices so that readers can inspect the study materials directly.

The main question was:

When the same AI data-sharing notice is shown visually instead of as plain text, does that change how willing people say they are to share their session data?

The survey also collected trust, completeness, ease of use, preference, and written reasons for participants' choices. This paper reports a preliminary descriptive coding of the written responses; it does not treat that coding as a formal qualitative study. The main measure was each participant's visual willingness score minus their plain-text willingness score. A positive score favors the visual notice.

2. Related Work

Privacy notice research shows that content is only part of the problem. Timing, layout, context, and the needs of the reader also affect whether a notice is useful. Schaub et al. (2015) describe privacy notices as a design problem, not just a legal document. This study follows that idea by holding the notice content steady and changing only its presentation.

Kelley et al. (2010) studied a privacy “nutrition label,” a structured format meant to make privacy practices easier to compare. The OptBot visual notice uses the same general idea: organize the information so people can scan it more easily. The visual version does not make stronger privacy claims than the plain-text version.

Egelman et al. (2009) found that the timing and placement of privacy information can affect decisions. For that reason, participants rated each notice right after seeing it. Privacy decisions are also shaped by attention, context, and differences in how people read policies (Acquisti et al., 2015; Reidenberg et al., 2015). Those findings support using both ratings and open-ended questions.

3. Method

3.1 Study design and materials

This was a paired study: every participant saw both notices.

- **Notice A:** a fixed icon-led visual disclosure ledger.
- **Notice B:** the same information in plain text.

The survey randomly chose which notice appeared first. It did not randomly assign people to a visual or plain-text-only group. This lets the study compare each person's two ratings and check whether order may matter. It does not prove that visual notices cause a change in willingness for all users.

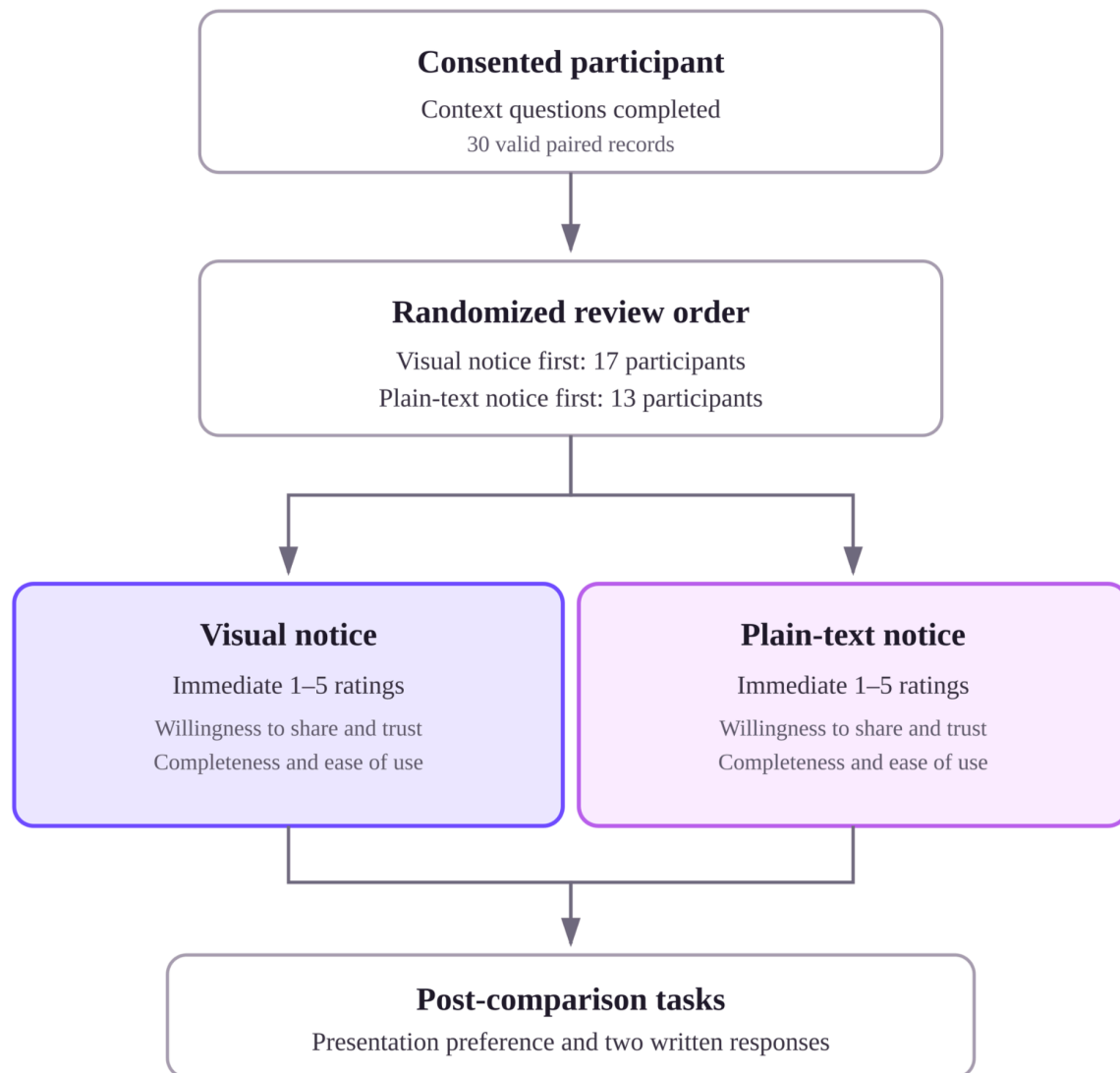


Figure 1. Each participant reviewed and rated both notices. Only the review order was randomized.

Both notices explained what data could be shared, how it could be used, privacy safeguards and their limits, and retention and user controls. After each notice, participants rated willingness to share, trust, completeness, and ease of use on a 1–5 scale. After both notices, they picked a preferred format and answered two short questions about the notices and their decision. Appendices A–C provide the exact administered wording and the distinct visual and plain-text presentations.

3.2 Building the survey in increments

The team improved the survey in steps before the v1.0.0 launch. Developers reviewed problems in earlier versions and made focused changes. The public repository records these updates (Candela, 2026).

Step	Change	Why it mattered
Clear comparison	Reduced several visual versions to one visual notice and one matched text notice	Kept the main comparison simple
Random order	Added random notice order and stable Notice A/B labels	Helped reduce first-notice bias
More realistic content	Replaced generic consent copy with concrete sharing, use, safeguards, retention, and control details	Made the task closer to a real data-sharing choice
Paired ratings	Added the same ratings after each notice and made willingness the main outcome	Made within-person comparisons possible
Submission checks	Required consent, review acknowledgements, complete ratings, and two written responses of at least five words	Improved data quality

3.3 Data handling and analysis

The study kept ratings separate from open-ended responses. This paper reports aggregate rating results and a preliminary descriptive coding of the open-ended responses. The qualitative coding identifies recurring design considerations; it is not a formal qualitative study or an inferential test.

A valid quantitative response needed consent, required answers, review acknowledgements, all eight ratings, valid timestamps, and two written responses of at least five words. The final quantitative data set had 30 valid paired responses, no excluded records, no duplicate response IDs, and no missing ratings. The qualitative-coding exclusions described in Section 5.1 apply only to interpretive analysis of the open-ended text.

The analysis looked at each participant's difference between the two notices. It reports the average and middle difference, the full difference pattern, and a 95% bootstrap confidence interval. The bootstrap used 100,000 resamples. The exact sign test checked whether positive and negative differences were balanced after ties were removed. Trust, completeness, and ease of use were extra exploratory outcomes, so the paper does not make a broad significance claim from them.

3.4 Recruitment and participant characteristics

Participants were recruited through the researchers’ convenience networks: friends, family members, colleagues, and classmates. Eligibility required participants to be at least 18 years old, to provide affirmative consent, and to complete the survey. The survey did not request direct personal identifiers; open-ended responses were retained only in the restricted workspace.

Thirty responses met the study’s completion and quality rules. Participation was uncompensated. Table 2 and Figure 2 report the collected age-range and AI-tool-use distributions from the approved quantitative snapshot.

3.5 Follow-on recruitment plan

A larger study should recruit about 120 adults through university, community, and online channels, with consent and modest compensation. It should include people across age groups and AI-use levels. Before collecting data, the team should set the sample target, main outcome, exclusion rules, and analysis plan in advance.

4. Results

4.1 Final sample

Measure	Value
Valid paired responses	30
Excluded or quality records	0
Missing paired ratings	0
Visual notice first	17
Plain-text notice first	13

Table 1. Final sample and data-quality checks.

Characteristic	Category	Count	Percentage
Age range	18–24	14	46.7%
	25–34	10	33.3%
	35–44	2	6.7%
	45–54	0	0.0%
	55–65	4	13.3%
	Prefer not to say	0	0.0%
AI-tool-use frequency	Never or almost never	6	20.0%
	Monthly	6	20.0%
	Weekly	8	26.7%
	Daily	10	33.3%

Table 2. Participant characteristics

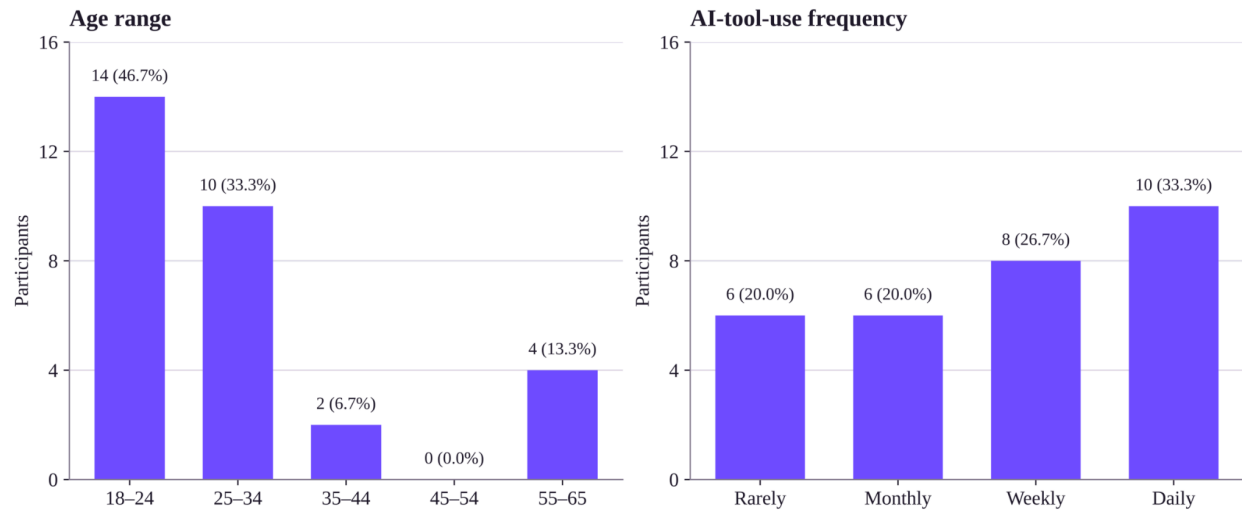


Figure 2. Age range and AI-tool-use frequency among valid paired responses. The percentage axes start at zero and each bar reports both count and percentage.

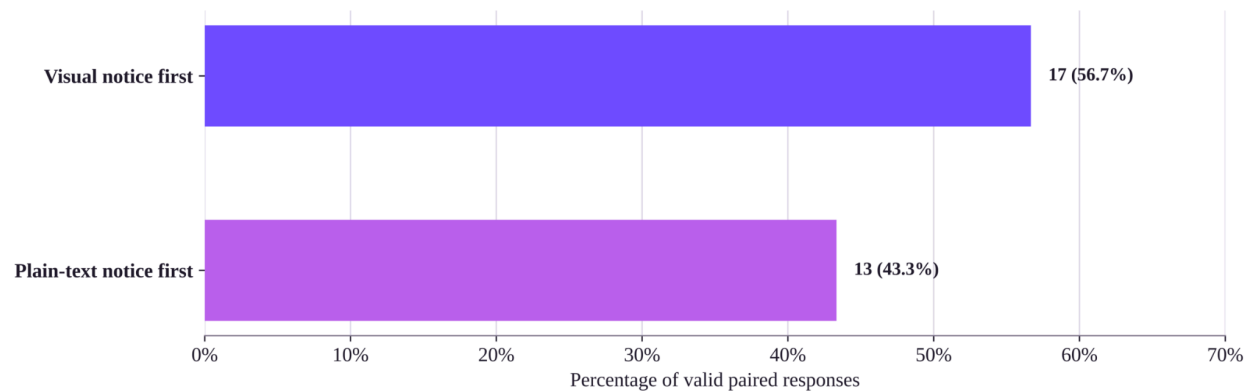


Figure 3. The order chart starts at zero and reports both counts and percentages of the 30 valid paired responses.

4.2 Main result: willingness to share

The visual notice had an average willingness score of 3.40. Plain text had an average of 3.20. The average difference was +0.20, but the confidence interval crossed zero. Sixteen of 30 participants gave the same willingness rating to both notices. The exact sign test was not statistically significant ($p = .424$).

Statistic	Value
Visual mean	3.40
Plain-text mean	3.20
Average paired difference	+0.20
Middle paired difference (median)	0
95% bootstrap confidence interval	[−0.07, 0.50]
Higher visual / higher text / tied	9 / 5 / 16
Exact two-sided sign test	$p = .424$

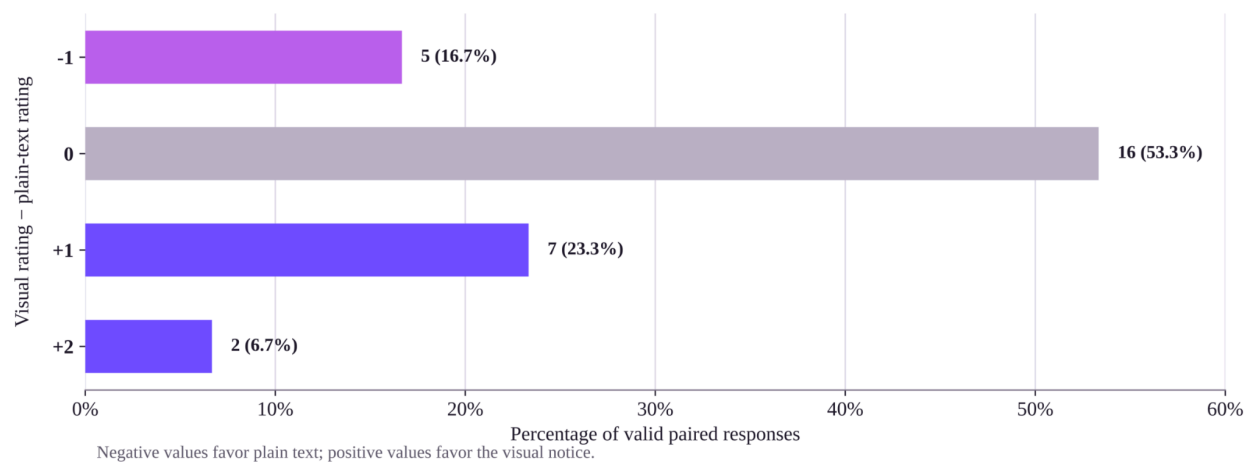


Figure 4. Most participants gave tied willingness ratings. The zero-based chart reports the full paired pattern as both counts and percentages, not just the two averages.

The main result is uncertain. The data could fit a small visual-notice benefit or no useful difference.

Willingness differences were +0.06 when the visual notice came first ($n = 17$) and +0.38 when plain text came first ($n = 13$). These groups are small. The numbers simply show why the review order was recorded; they do not prove an order effect.

4.3 Paired rating distributions

The restricted working file supplied for qualitative review, contains only open-ended responses. The approved quantitative snapshot supplied aggregate rating distributions for all eight rating fields.

Figure 5 reports the count and percentage at each rating for each notice. The percentage axes begin at zero, and each distribution has $n = 30$.

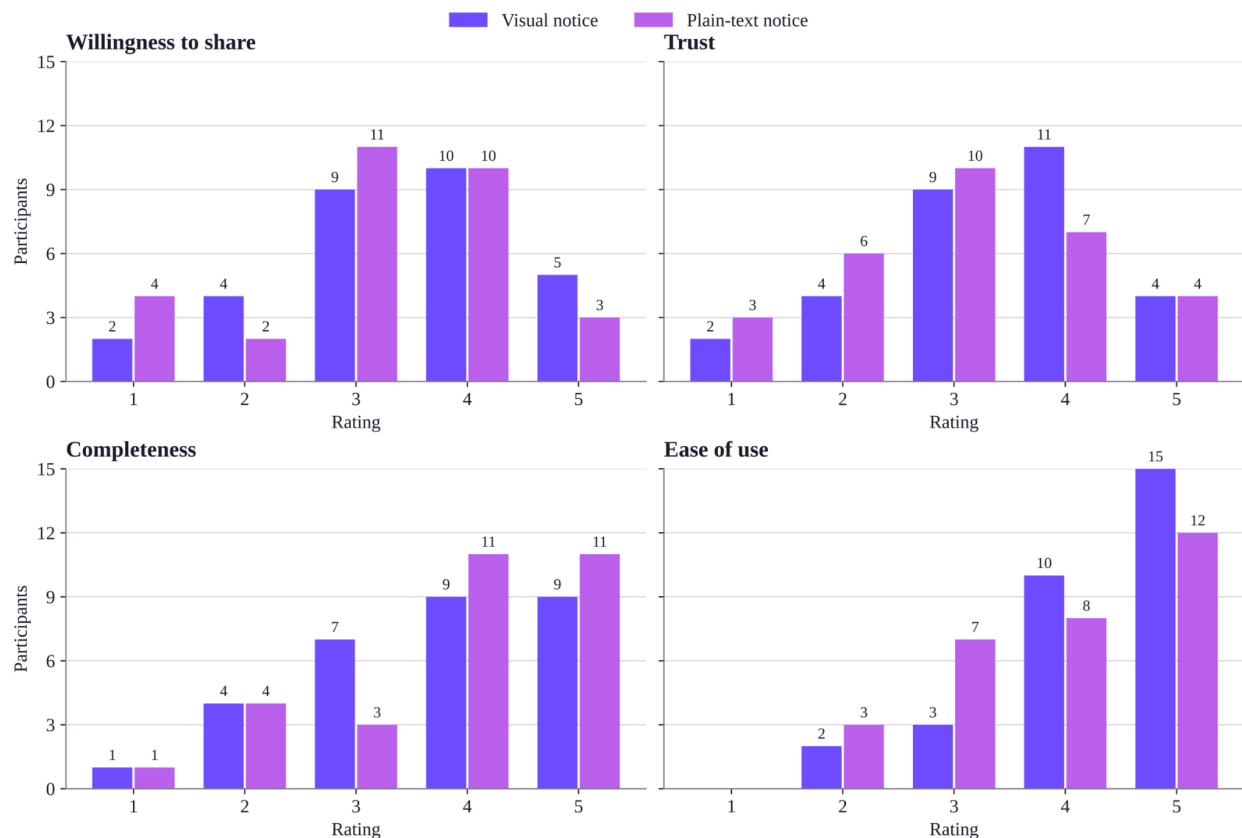


Figure 5. Counts and percentages for each 1–5 rating. Trust, completeness, and ease of use are exploratory outcomes.

Figure 6 adds boxplots as a secondary summary of these marginal rating distributions. The boxplots do not show the within-participant links between Notice A and Notice B; the paired willingness-difference distribution remains in Figure 4. For the final analysis, a paired dot-and-line plot would be the preferred complement if the authorized quantitative workspace can produce it.

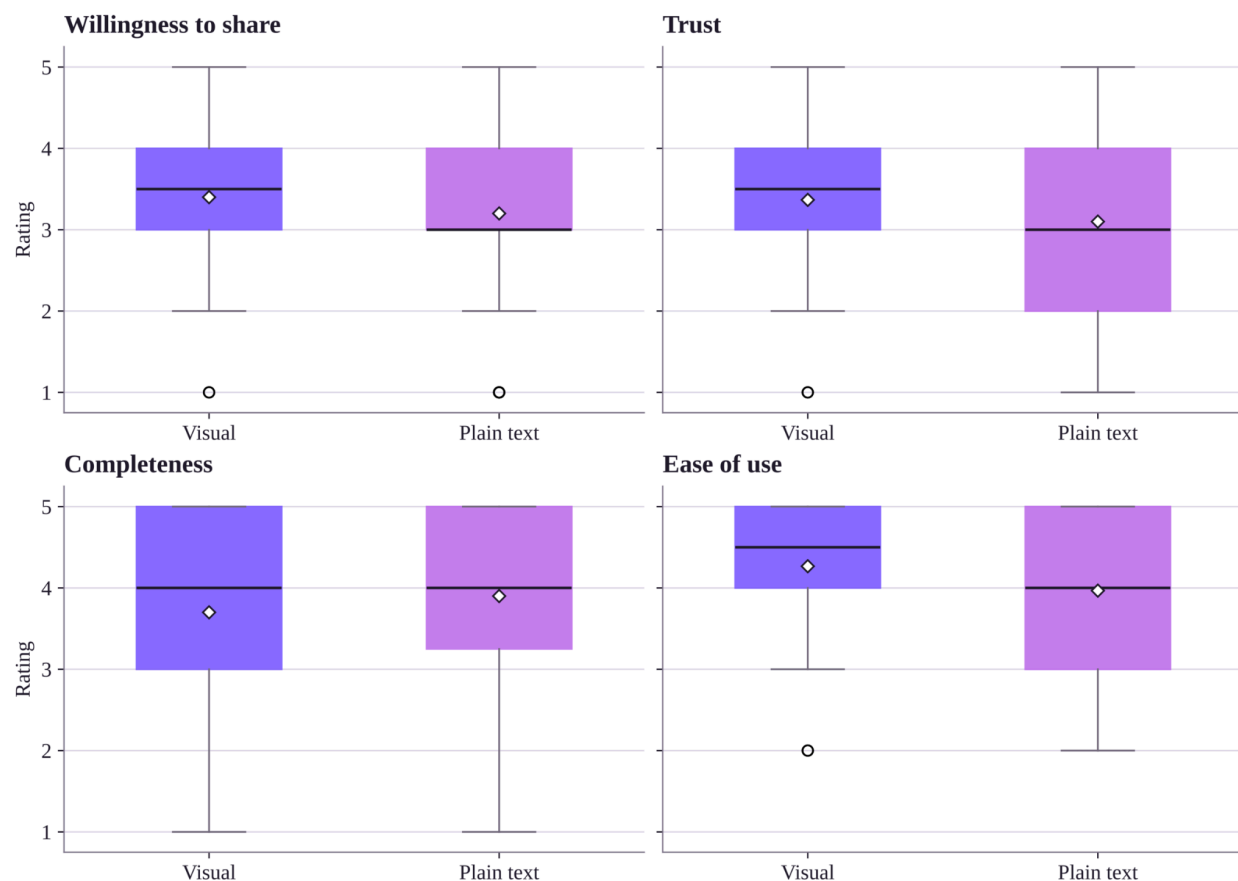


Figure 6. Boxplots summarize the aggregate 1–5 rating distributions for each notice. They are descriptive and do not replace paired analysis.

4.4 Other outcomes and preference

Outcome	Visual mean	Plain-text mean	Average difference	95% bootstrap CI	Exact sign test, unadjusted
Trust	3.37	3.10	+0.27	[−0.03, 0.60]	.332
Completeness	3.70	3.90	−0.20	[−0.53, 0.13]	.454
Ease of use	4.27	3.97	+0.30	[−0.07, 0.67]	.092

Nineteen participants preferred the visual notice and 11 preferred plain text (63.3% vs. 36.7%). Preference was measured after both notices, so it is useful for design ideas but is not separate proof that one format caused a change.

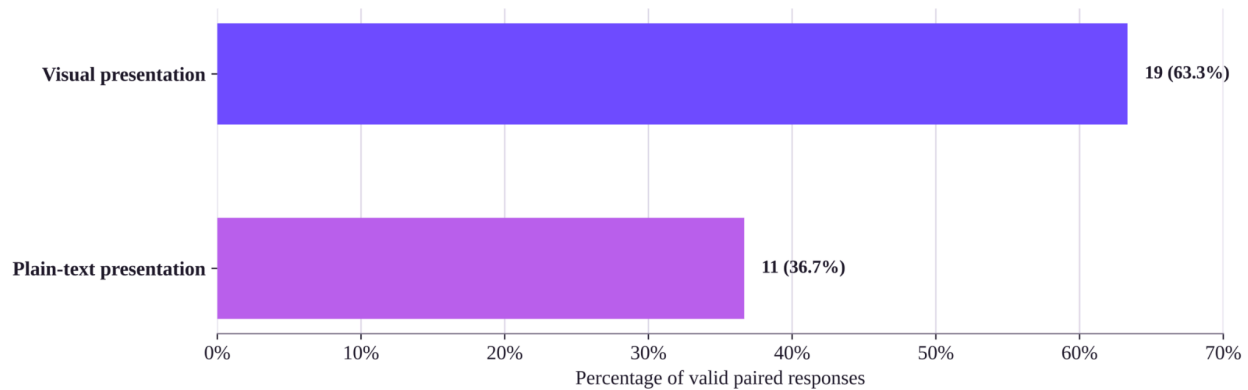


Figure 7. The zero-based chart reports both counts and percentages. Preference is descriptive because it was measured after both notice reviews.

5. Discussion

The study did not find a statistically significant difference in willingness to share. The visual notice had a small positive average difference, but the median paired difference was zero and the confidence interval included zero. This does not prove that there is no effect. It means this small sample cannot clearly separate a small effect from no effect.

The other ratings point to possible design questions. The visual notice had higher average trust and ease-of-use ratings. Plain text had a higher average completeness rating. One explanation is that visual grouping may help people scan a notice, while denser text may look more complete. Because all of these confidence intervals included zero, these patterns should be interpreted cautiously and tested in a larger study.

A strength of the project is the study tool and process. The final survey used matched content, random review order, paired ratings, response checks, and separate handling of open-ended response text. These choices make the project easier to review and improve than a survey that changes several things at once.

5.1 Preliminary qualitative observations and design implications

The two written prompts elicited 30 response pairs. A rapid descriptive coding treated themes as overlapping: a response could indicate more than one theme. One response pair containing placeholder text was excluded from all qualitative coding. Two additional decision-influence responses did not state a substantive rationale and were not assigned decision-influence themes. The resulting counts are descriptive indicators, not prevalence estimates or tests of association.

Recurring theme	Response pairs indicating the theme (<i>n</i> = 29)	What participants indicated	Design implication
Specific data-use and control information	14	Decisions turned on what data would be shared, how it would be used, retention, deletion, and access consequences	Put collection, purpose, retention, deletion, and the option to decline in the first visible notice layer
Visual hierarchy and scanability	11	Participants described icons, spacing, grouping, and concise language as easier to scan or understand	Keep the four-part visual structure, but pair each icon with a descriptive heading and plain-language text
Pre-existing AI and privacy skepticism	11	Some participants said that prior concerns about AI safety or data sharing outweighed notice formatting	Do not frame visual polish as a substitute for privacy protections; state safeguards, their limits, and user controls concretely
Detail, density, and perceived completeness	10	The visual version could feel simplified, while plain text could feel more complete but denser	Use progressive disclosure: retain the scannable summary and offer an immediate, clearly labeled full-detail view
Tone and perceived trustworthiness	8	Participants associated visual tone with friendliness or trust, but some read it as overly persuasive; plain text could feel formal or robotic	Use neutral, direct language and avoid decorative cues that could make a consent request feel promotional

Clarity and readability were recurring considerations in 11 of the 29 coded response pairs. Participants often described Notice A as easier to scan because of its icons, spacing, grouping, and concise sections. Other participants described Notice B as more detailed, formal, or technically complete. The responses

were mixed: some participants viewed Notice A as clear and engaging, while others viewed it as overly simplified or difficult to interpret. The qualitative evidence therefore does not identify one uniformly more readable format.

A related theme, present in 10 response pairs, was a perceived tradeoff between simplicity and completeness. Notices A and B contained the same disclosure terms in the same order, but some participants interpreted the visual presentation as a simplified version and the plain-text presentation as more complete. This suggests that visual structure can change perceived technical depth even when the underlying disclosure content does not change.

Visual hierarchy and scanability did not eliminate privacy concerns. In 11 response pairs, participants described pre-existing concerns about AI privacy, data collection, or trust that outweighed formatting preferences. Participants also focused on specific information about collection, use, retention, deletion, and user controls in 14 response pairs. These observations support a follow-up design that combines a scan-friendly visual summary with an immediately available detailed disclosure. The qualitative findings identify hypotheses for later testing; they do not show that visual presentation changed willingness to share.

6. Limits and Future Work

The sample was small and likely based on convenience. Everyone saw both notices, so seeing the first notice could affect reactions to the second. The visual notice also changed several features together, including icons, spacing, hierarchy, and density. The study cannot say which visual feature mattered. Finally, willingness is a self-report and the scenario was simulated.

A follow-up study should use a larger preregistered sample and either randomly assign people to one format or use stronger counterbalancing. It should add comprehension questions and an ethical simulated sharing choice. A qualitative follow-up should use two authorized coders, a shared codebook, agreement checks, and privacy review before sharing themes or quotes.

7. Conclusion

This study compared a visual and plain-text notice for a simulated AI model-improvement workflow. The visual notice had a small positive mean difference in willingness, but the confidence interval included zero and the sign test was not statistically significant. The best conclusion is cautious: the design is worth studying further, but this sample does not justify a strong claim about visual notices increasing consent.

The OptBot survey tool provides a useful starting point for future usable-privacy work. It uses matched notices, random order, paired ratings, validation checks, and separate handling of qualitative data. A larger study can test whether specific design choices improve understanding or affect privacy-related behavior.

References

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Appendix A. Survey as administered

This appendix transcribes the survey flow and the consent screen. The wording below is the wording presented to participants.

A.1 Welcome screen

Title: Review two privacy notices for a simulated OptBot Assistant service.

OptBot Assistant is a simulated artificial intelligence (AI) service. This study asks how people respond to two presentations of the same model-improvement notice.

- Focus: Model-improvement privacy notices
- Time: About 3–5 minutes
- Data: Survey answers

A.2 Consent screen

Title: Confirm that you want to participate.

Participation is voluntary. Please confirm that you are at least 18 years old and agree to answer this study survey.

- I agree to participate — Continue to the study questions.
- I do not agree — Exit without submitting a response.

A.3 Participant-context questions

Title: Share a little context before reviewing the notices.

These two questions provide basic context without collecting direct personal identifiers.

1. **Age range:** 18–24; 25–34; 35–44; 45–54; 55–65; or Prefer not to say.
2. **How often do you use AI tools?** Never or almost never; Monthly; Weekly; or Daily.

A.4 Review instructions

Title: Compare two presentations of the same privacy notice.

Notice A and Notice B contain the same terms in the same order but differ in presentation. Their review order is set for this session.

- **Review a notice:** Read what is shared, how OptBot uses it, and how long OptBot keeps it.
- **Rate it:** Answer the same four questions immediately after each notice.
- **Compare both:** Choose the presentation you prefer, then briefly explain your decision.

A.5 Notice review and ratings

Participants reviewed either Notice A or Notice B first according to the randomized review order, selected the matching “I reviewed Notice [A/B]” acknowledgement, and answered the following items on a 1–5 scale. They then reviewed and rated the other notice using the same items, scale, and anchors.

Item	Low anchor	High anchor
I would be willing to share my session data under this notice.	Not willing	Very willing
This notice made the AI system feel trustworthy.	Low trust	High trust
This notice included enough information for me to make a decision.	Incomplete	Complete
This notice was easy to scan and use.	Difficult	Easy

A.6 Preference and written responses

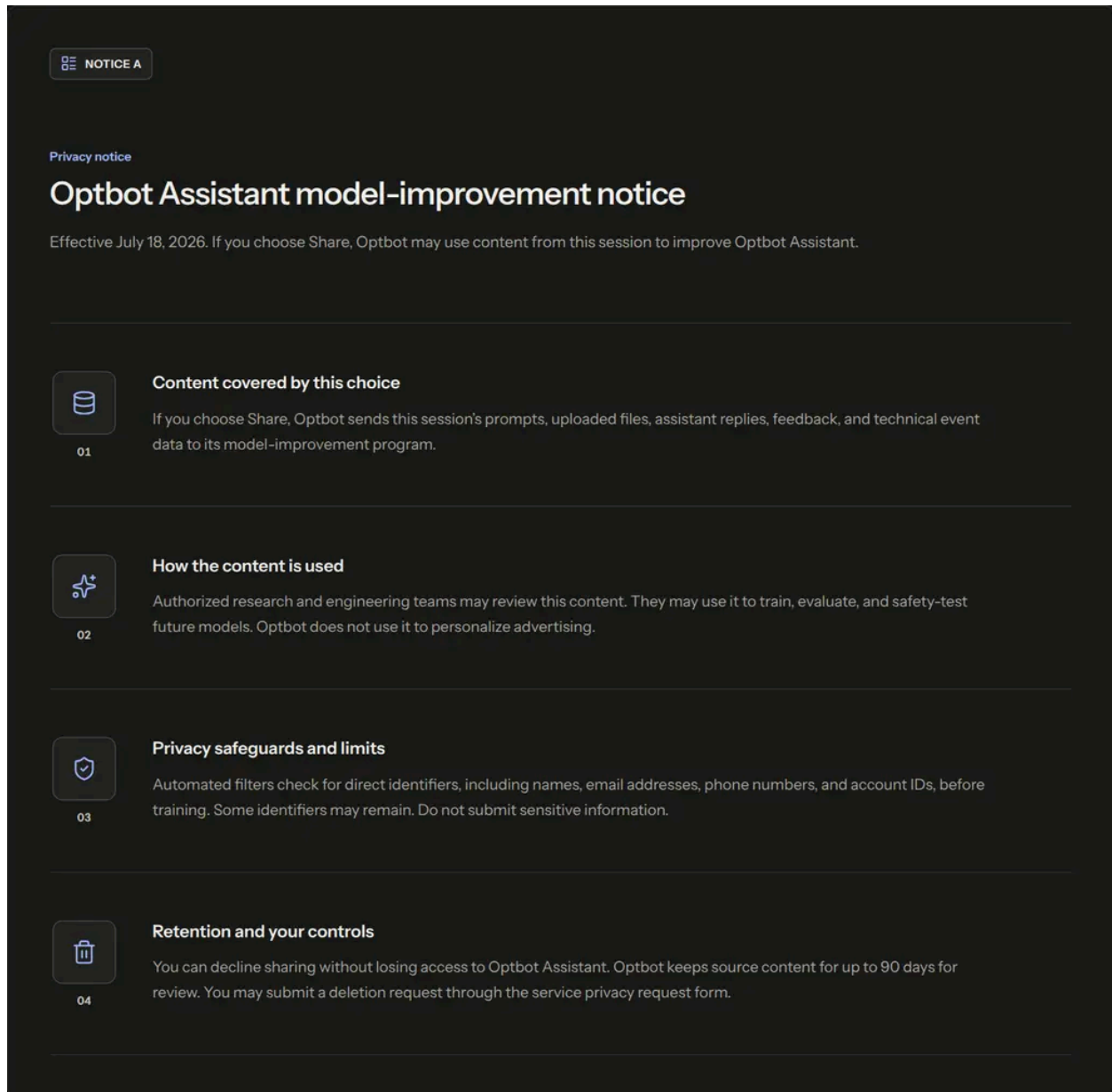
Preference prompt: Which notice would you prefer before deciding whether to share data?

- Visual presentation
- Plain-text presentation

Written-response instruction: Use your own words. Answer both questions in at least five words each.

1. What words or short phrases would you use to describe Notice A and Notice B?
2. What most influenced your willingness or unwillingness to share?

Appendix B. Accessible transcription of Notice A: Icon-led visual disclosure



OptBot Assistant model-improvement notice

Effective July 18, 2026. If you choose Share, OptBot may use content from this session to improve OptBot Assistant.

The image above reproduces the Notice A presentation shown to participants. The accessible transcription below preserves its content, order, and visual cues.

Visual cue	Heading	Disclosure text
Database icon · 01	Content covered by this choice	If you choose Share, OptBot sends this session's prompts, uploaded files, assistant replies, feedback, and technical event data to its model-improvement program.
Sparkles icon · 02	How the content is used	Authorized research and engineering teams may review this content. They may use it to train, evaluate, and safety-test future models. OptBot does not use it to personalize advertising.
Shield icon · 03	Privacy safeguards and limits	Automated filters check for direct identifiers, including names, email addresses, phone numbers, and account IDs, before training. Some identifiers may remain. Do not submit sensitive information.
Trash icon · 04	Retention and your controls	You can decline sharing without losing access to OptBot Assistant. OptBot keeps source content for up to 90 days for review. You may submit a deletion request through the service privacy request form.
<i>The rendered image and the transcription identify the four icons, numbered sections, order, headings, and disclosure text used in Notice A.</i>		

Appendix C. Notice B: Plain-text disclosure as administered

 NOTICE B

Optbot Assistant model-improvement notice

Effective July 18, 2026. If you choose Share, Optbot may use content from this session to improve Optbot Assistant.

Content covered by this choice

If you choose Share, Optbot sends this session's prompts, uploaded files, assistant replies, feedback, and technical event data to its model-improvement program.

How the content is used

Authorized research and engineering teams may review this content. They may use it to train, evaluate, and safety-test future models. Optbot does not use it to personalize advertising.

Privacy safeguards and limits

Automated filters check for direct identifiers, including names, email addresses, phone numbers, and account IDs, before training. Some identifiers may remain. Do not submit sensitive information.

Retention and your controls

You can decline sharing without losing access to Optbot Assistant. Optbot keeps source content for up to 90 days for review. You may submit a deletion request through the service privacy request form.

OptBot Assistant model-improvement notice

Effective July 18, 2026. If you choose Share, OptBot may use content from this session to improve OptBot Assistant.

Notice B used the same headings and disclosure text in the same order, but without the numbered icon ledger used in Notice A.

Content covered by this choice

If you choose Share, OptBot sends this session's prompts, uploaded files, assistant replies, feedback, and technical event data to its model-improvement program.

How the content is used

Authorized research and engineering teams may review this content. They may use it to train, evaluate, and safety-test future models. OptBot does not use it to personalize advertising.

Privacy safeguards and limits

Automated filters check for direct identifiers, including names, email addresses, phone numbers, and account IDs, before training. Some identifiers may remain. Do not submit sensitive information.

Retention and your controls

You can decline sharing without losing access to OptBot Assistant. OptBot keeps source content for up to 90 days for review. You may submit a deletion request through the service privacy request form.